



Introduction to Mechatronics

ME 350: Design & Manufacturing II

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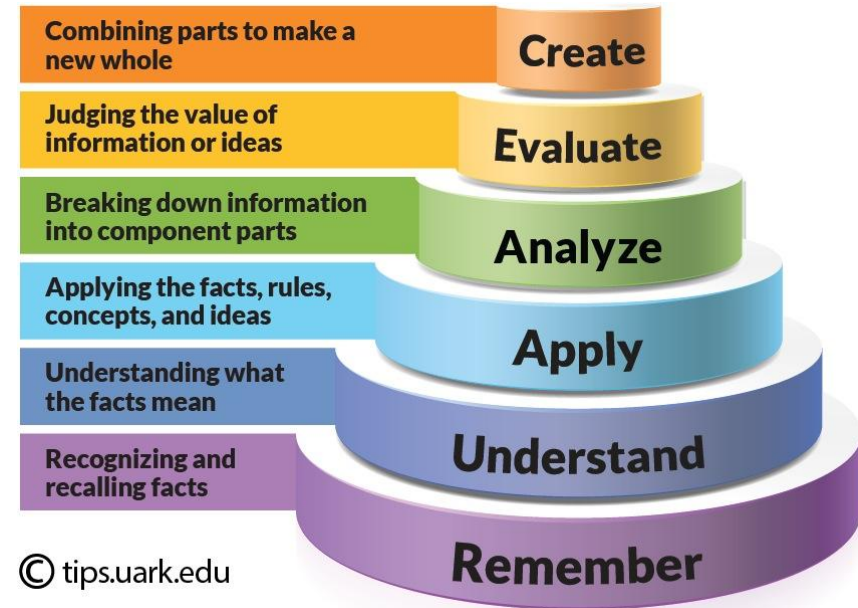
Some slides adopted from Mike Umbriac, David Remy, Chinedum Okwudire, & Shorya Awtar



Today's Learning Objectives

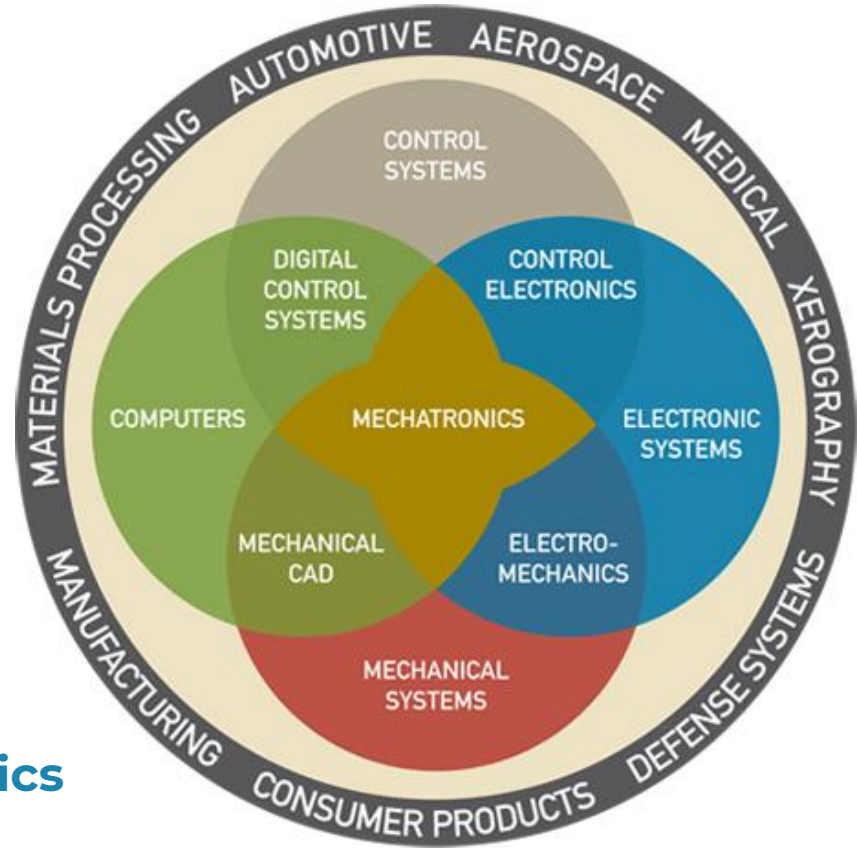


- Understand the definition of mechatronics and the basic elements of mechatronic systems
- Understand why mechanical engineers are often best positioned to take a lead role in the design of mechatronic systems
- Understand the conceptual design of a mechatronic system and how it applies to your ME 350 project



What is Mechatronics?

Mechatronics is the *synergistic* integration of mechanical disciplines, electronics, controls, and computers in the design of high performance systems.



Mechatronics = **Mechan**ics + Elect**ronic**s

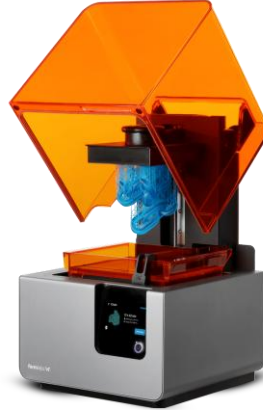
Mechatronic Systems in your Daily Life



Given this introductory definition to what a mechatronic system is, take a minute to think through your day until this point.

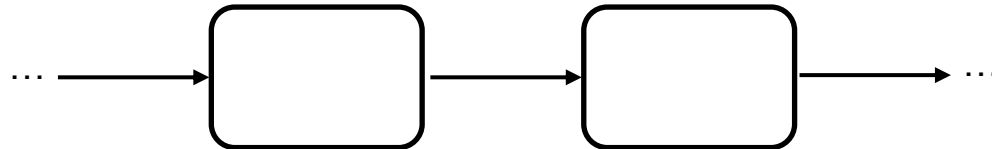
- What mechatronic systems have you interfaced with today?
- How many?
- Talk through it with the person next to you.

Some Examples of Mechatronic Systems

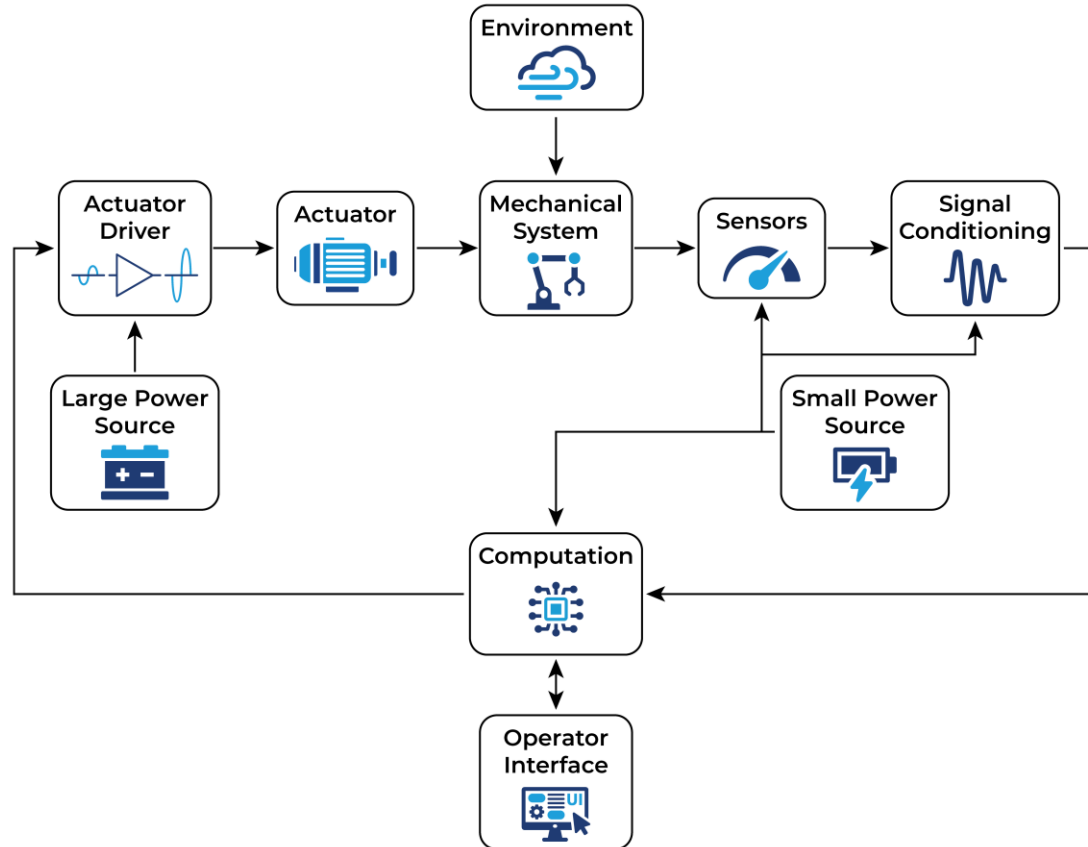


Representing Mechatronic Systems

- Block Diagrams
 - Each block represents a different element such as a sensor, power supply, actuator, controller, or the physical system itself (often referred to as the plant)
 - Between blocks are arrows that represent the flow of...
 - Information (signal, typically low current $<10\text{mA}$)
 - Power (typically large current $>100\text{ mA}$)
 - Modulated power (power that is directly proportional to the information signal)



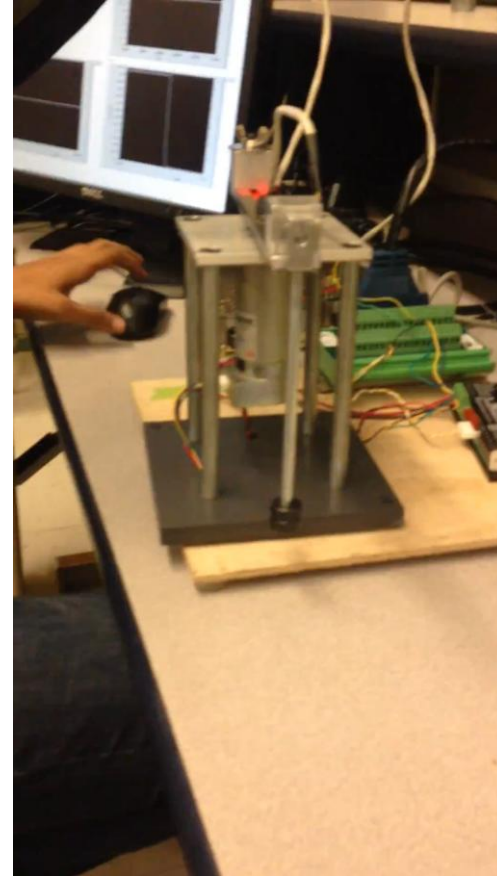
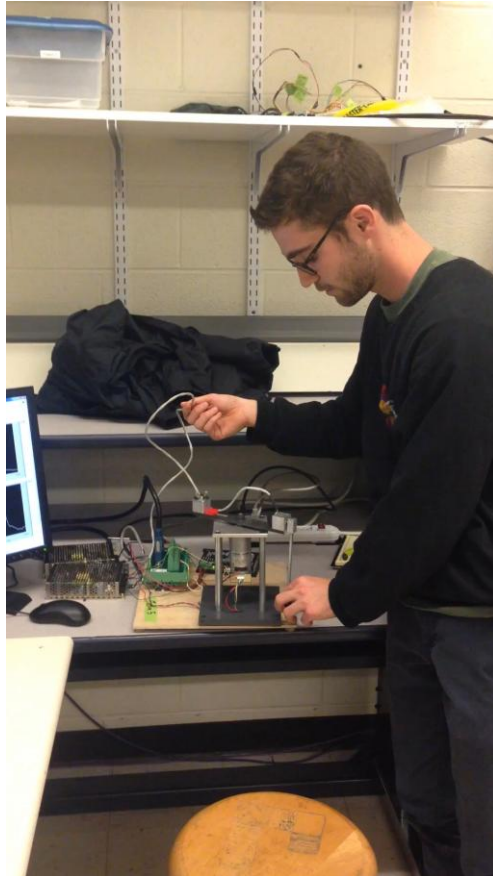
Block Diagram of a General Mechatronic System



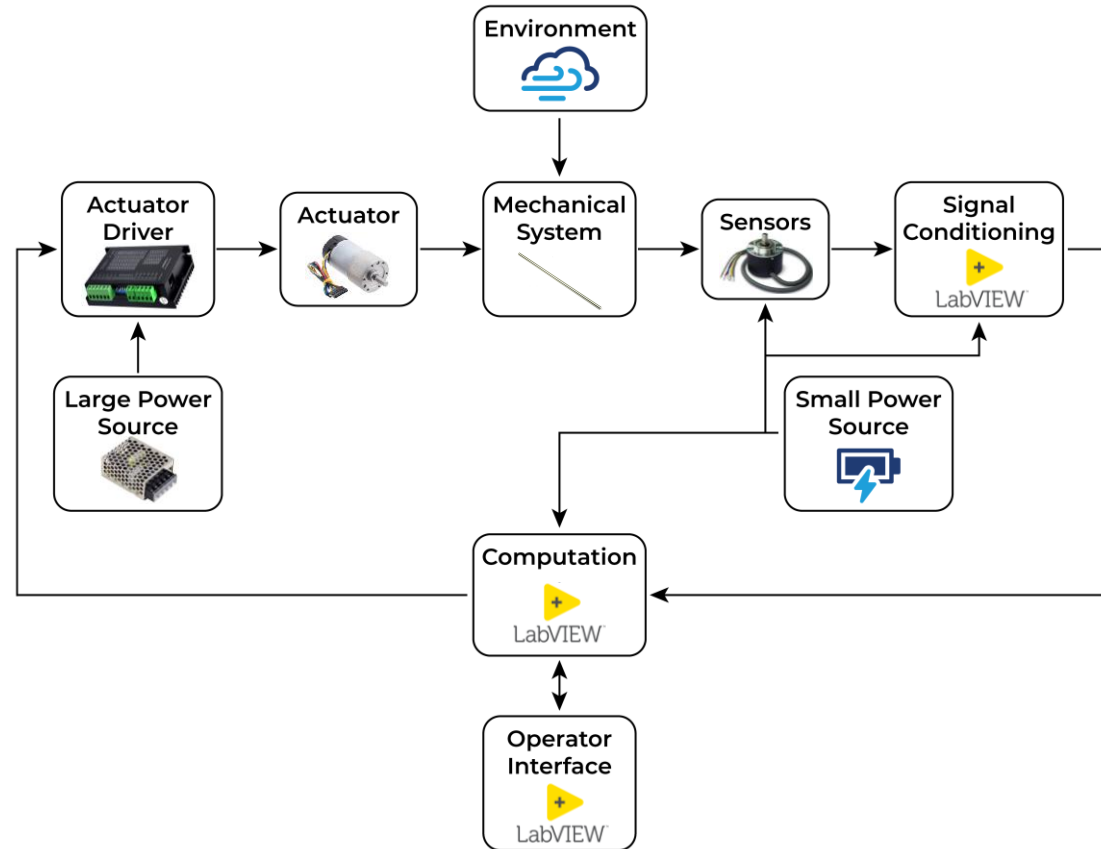
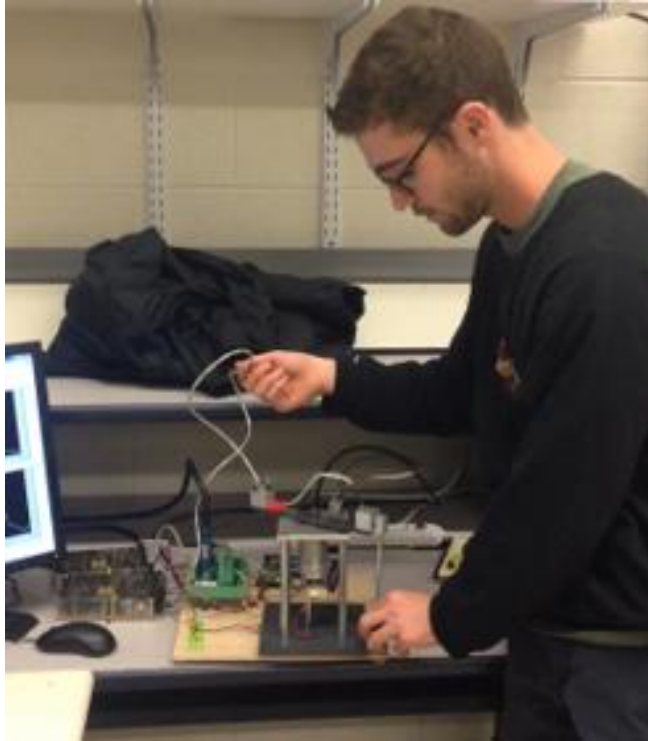
Case Study 1 – Simple Example



Case Study 1 – Inverted Pendulum



Case Study 1 – Inverted Pendulum



Case Study 1 – Inverted Pendulum

- Why care?



Case Study 2 – Modern Cars

1886 Benz Patent-Motowagen

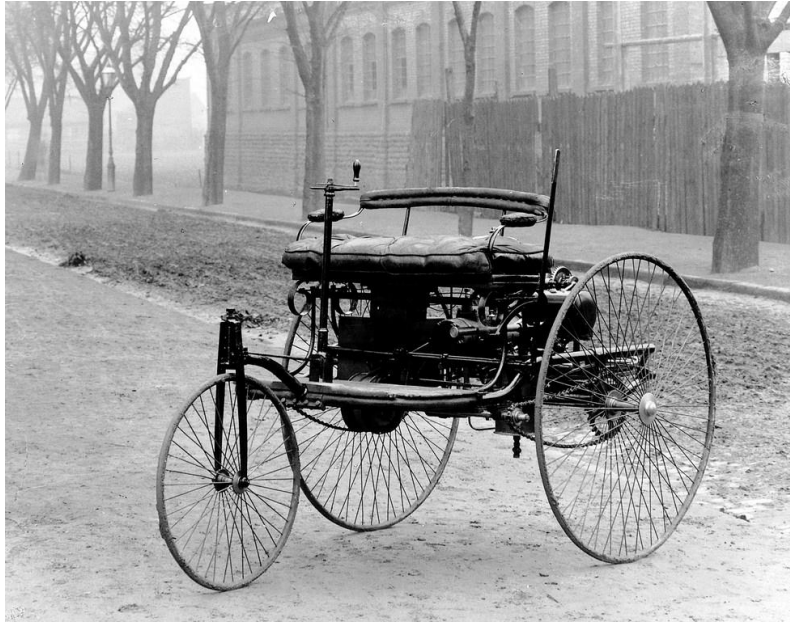


2019 Subaru Outback



Case Study 2 – Modern Cars

1886 Benz Patent-Motowagen



- Power train
 - 0 Sensors
- Suspension
 - 0 Sensors
- Safety
 - 0 Sensors
- Comfort
 - 0 Sensors

Total: 0 Sensors, 0 ECUs

ECU = Electronic Control Unit (essentially a computer)

Case Study 2 – Modern Cars

- Power train
 - 15-20 Sensors (Oxygen sensors, fuel level, RPM, temperature, ...)
- Suspension
 - 10-15 Sensors (ABS, ESP, ...)
- Safety
 - 20-25 Sensors (Airbags, Eyesight, Belt Pretensioner, ...)
- Comfort
 - 10-20 Sensors (Climate Control, Windows, ...)

Total: 55-80 Sensors, 30-50 ECUs

2019 Subaru Outback



Case Study 2 – Modern Car Suspension

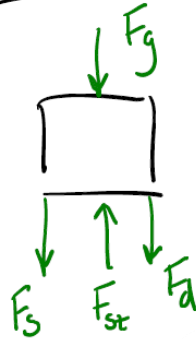
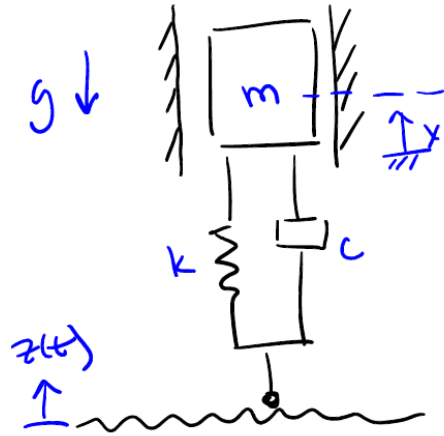


Case Study 2 – Modern Car Suspension

Quarter Car Model (Base Excitation)

Find: EOM & how the system responds to freq. inputs.

FBD



where $F_{st} = mg$ ← equilibrium at $x=0$

Newton's $m\ddot{x} = -k(x-z) - c(\dot{x}-\dot{z}) + \cancel{F_{st}} - \cancel{mg}$

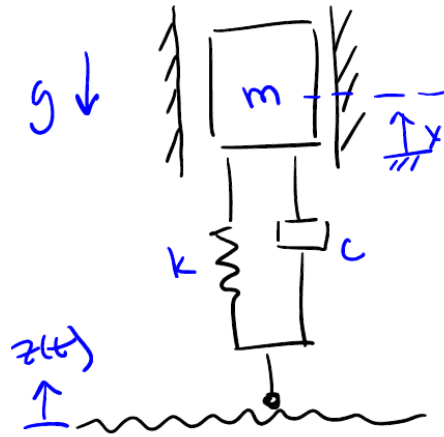
$$\ddot{X} + \underbrace{\frac{c}{m}\dot{X}}_{2\zeta\omega_n} + \underbrace{\frac{k}{m}X}_{\omega_n^2} = \underbrace{\frac{c}{m}\ddot{z}}_{\text{from left } 2\zeta\omega_n} + \underbrace{\frac{k}{m}z}_{\text{from left } \omega_n^2}$$

Assuming $z(t) = A \cos \omega t$

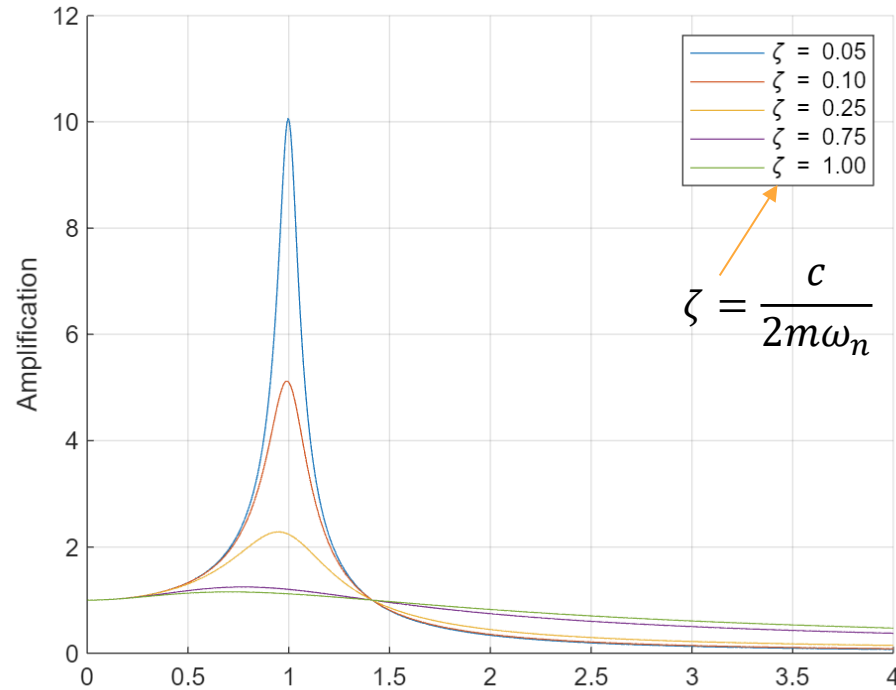
$$\omega_n = \sqrt{\frac{k}{m}} \quad \& \quad \zeta = \frac{c}{2m\omega_n}$$

Case Study 2 – Modern Car Suspension

Quarter Car Model



Assuming $z(t) = A \cos \omega t$



$$\zeta = \frac{c}{2m\omega_n}$$

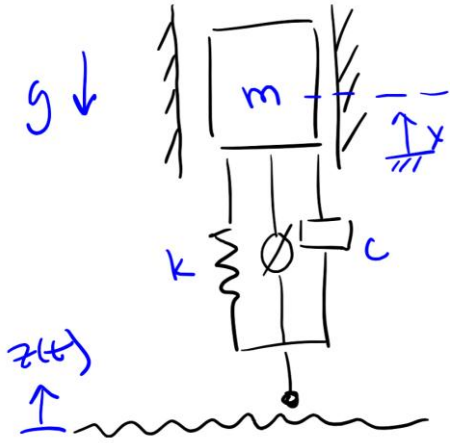
"Tuning Ratio" $\rightarrow \eta$

$$\eta = \frac{\omega}{\omega_n}$$

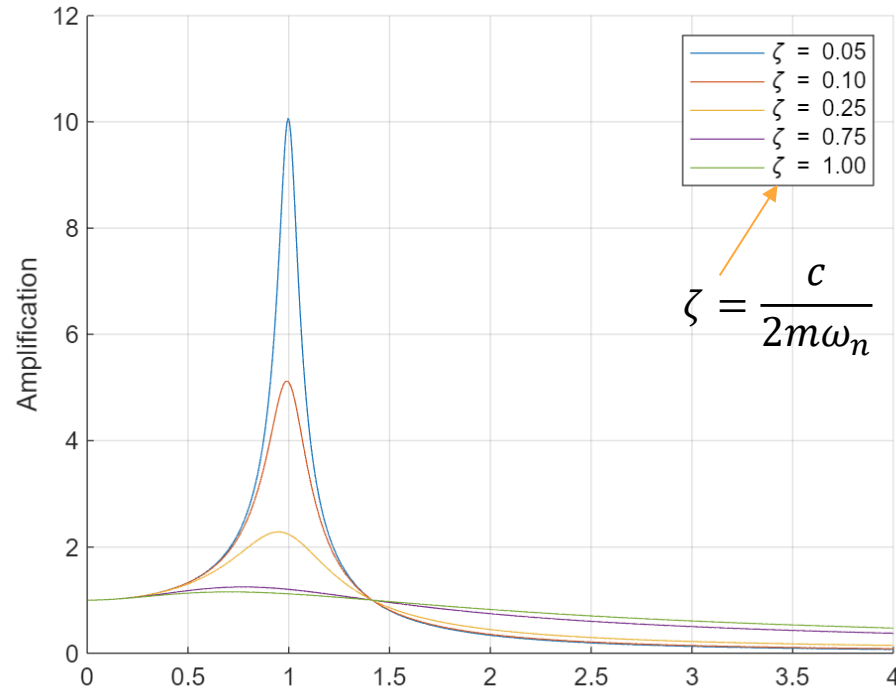
$$\omega_n = \sqrt{\frac{k}{m}}$$

Case Study 2 – Modern Car Suspension

Quarter Car Model



Assuming $z(t) = A \cos \omega t$



$$\zeta = \frac{c}{2m\omega_n}$$

"Tuning Ratio" $\rightarrow \eta$

$$\eta = \frac{\omega}{\omega_n}$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

Case Study 2 – Modern Car Suspension



Case Study 2 – Modern Cars Suspension



Take a Stretch Break

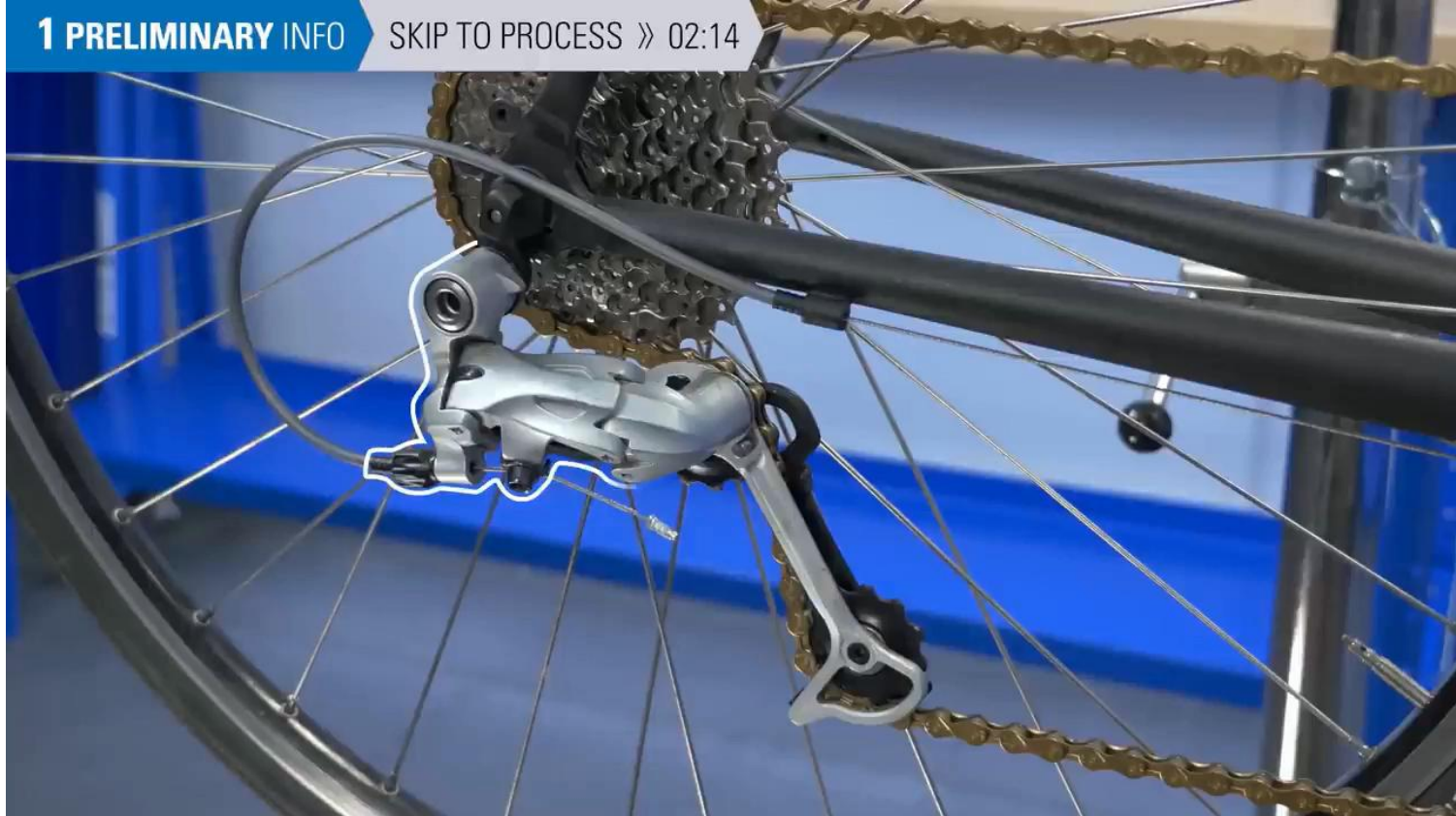


Case Study 3 – Bike Shifting

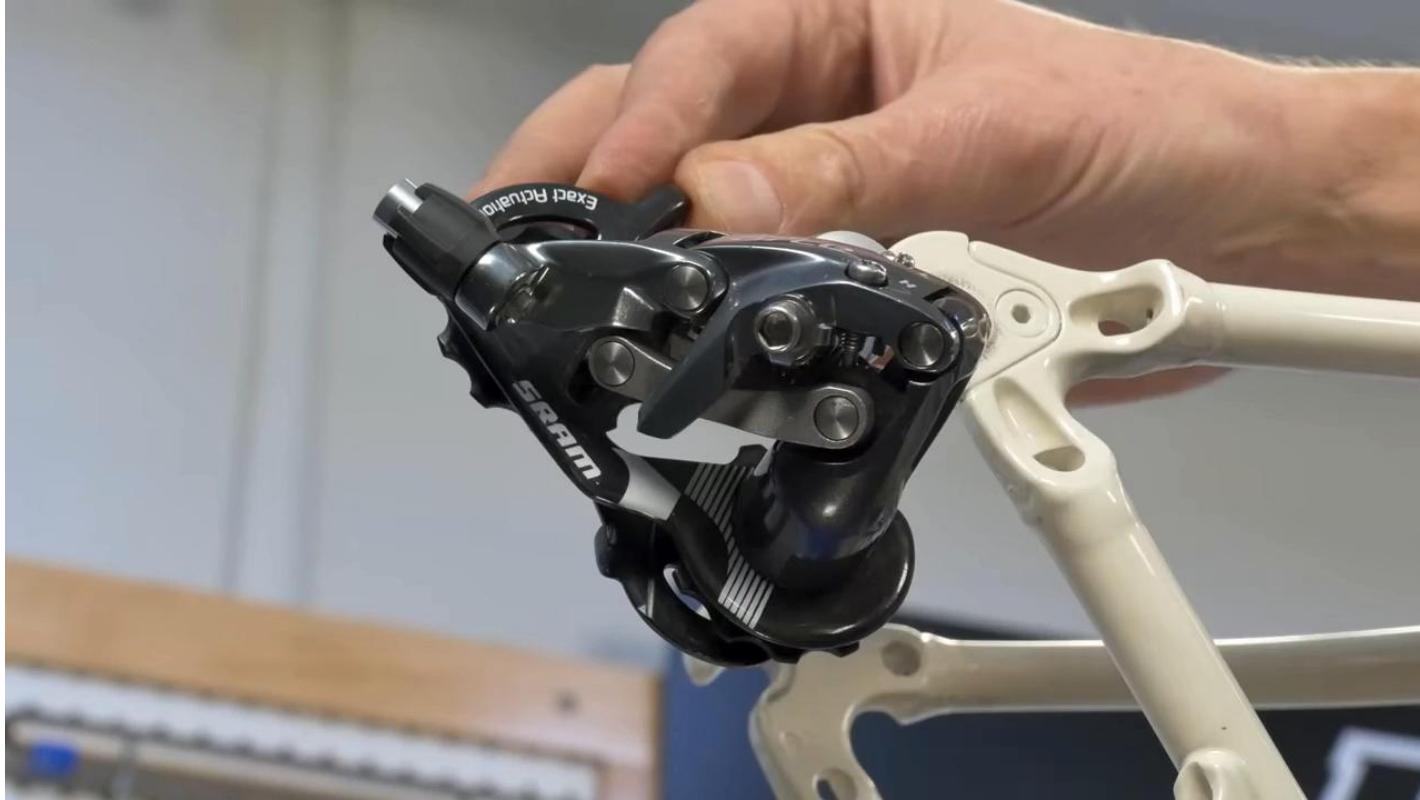


1 PRELIMINARY INFO

SKIP TO PROCESS » 02:14



Case Study 3 – Bike Shifting

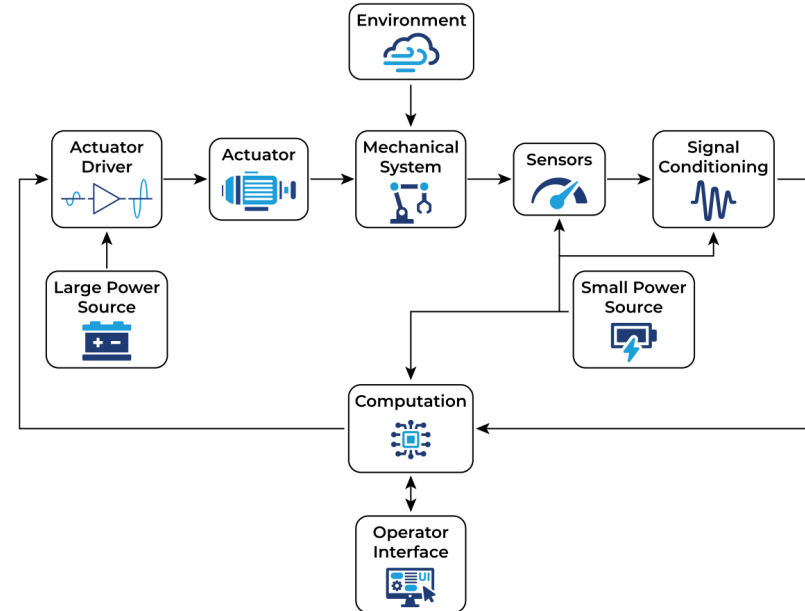


Case Study 3 – Bike Shifting



Case Study 3 – Bike Shifting

- What are some pros and cons of the electric shifting versus mechanical? Was this a necessary mechatronic intervention?



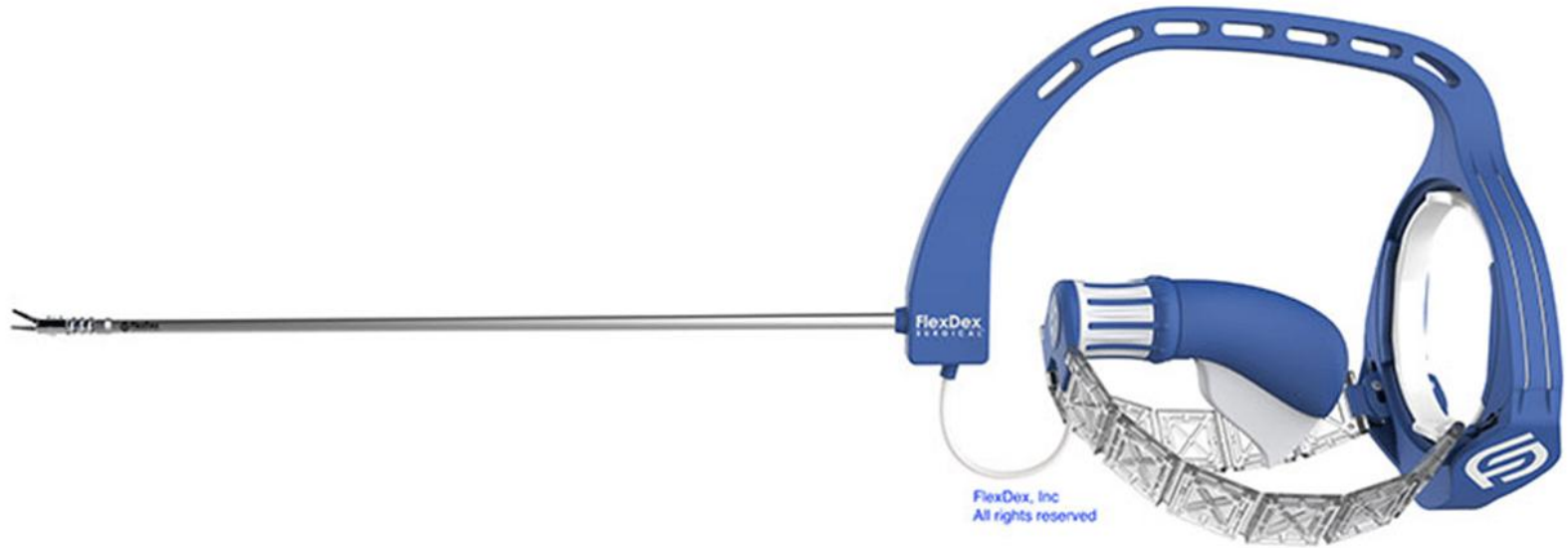
Case Study 4 – Surgical Robot



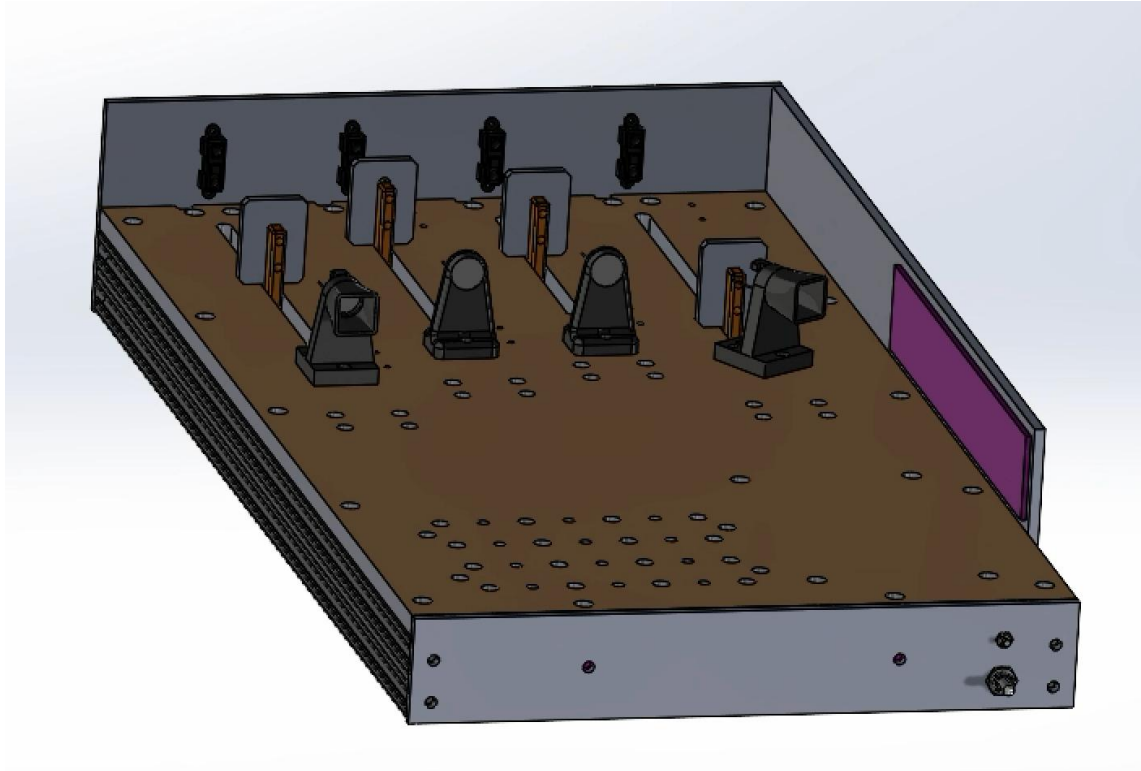
Case Study 4 – Surgical Robot



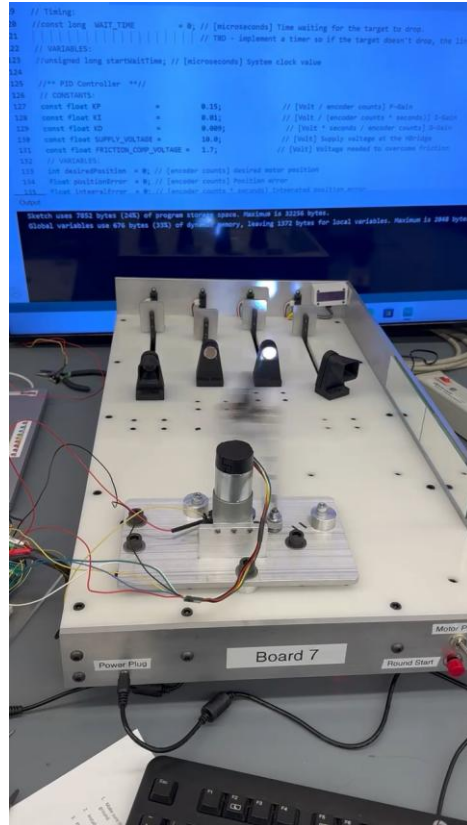
Case Study 4 – Surgical Robot



Mechatronic Components of Your ME 350 Project

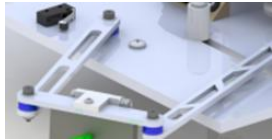


Mechatronic Components of Your ME 350 Project



Mechatronic Components of Your ME 350 Project

- Place the letter of each component next to the block(s) to which it corresponds



A



E



B



F



C



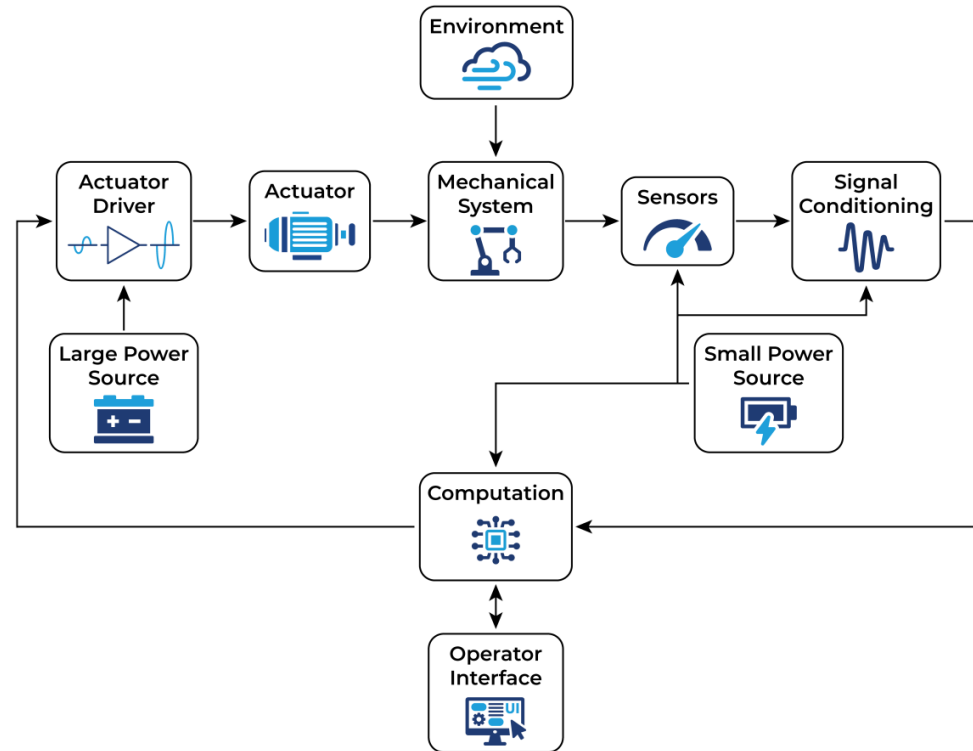
G



D

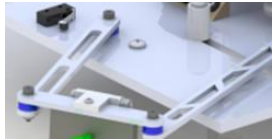


H



Mechatronic Components of Your ME 350 Project

- Place the letter of each component next to the block(s) to which it corresponds



A



E



B



F



C



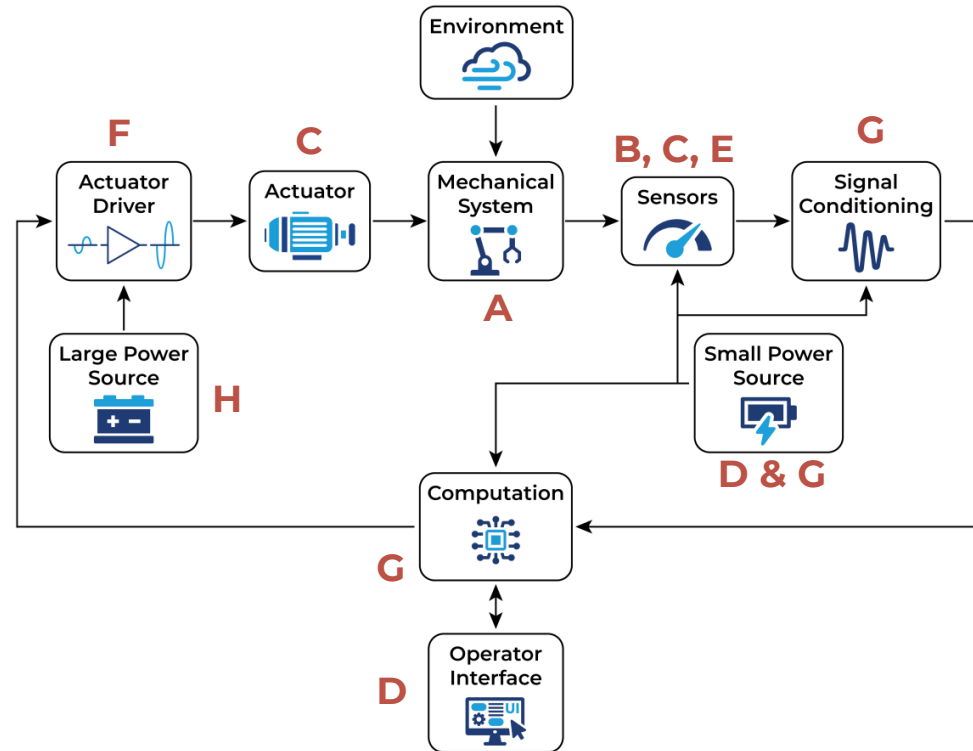
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D

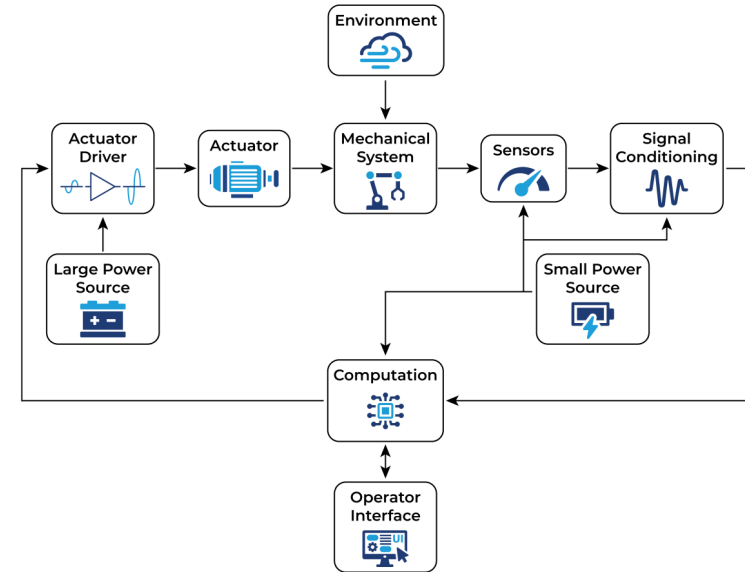


H



To Recap...

- Mechatronics is the synergistic integration of mechanical disciplines, electronics, controls, and computers in the design of high performance systems.
- There are many examples of mechatronic systems
 - Sometimes the mechatronic interventions are not necessarily the best option.



Moving Forward...

- We will start diving into individual aspects of a mechatronic system that we have yet to discuss in this course
- Next two classes we will look at sensors

